Representations of $\mathfrak{sp}_4\mathbb{C}$

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1 Basics of $\mathfrak{sp}_4\mathbb{C}$

Definition 1.1. (V, ω) is called a ***(complex) symplectic vector space*** if V is a complex vector space and ω is a non-degenerate skew-symmetric 2-form. A linear map $T : V \to V$ is called a ***symplectomorphism*** if $T^*\omega = \omega$. The set of all symplectomorphisms forms a Lie subgroup of $\mathrm{GL}(V)$ called the symplectic group $\mathrm{Sp}(V)$. The Lie algebra $\mathfrak{sp}(V)$ of $\mathrm{Sp}(V)$ is called the symplectic Lie algebra of V.

Suppose $\dim_{\mathbb{C}} V = 4$. By linear algebra, one can choose a symplectic basis of (V, ω), i.e., a basis $\{e_1, e_2, f_1, f_3\}$ such that

$$\omega(e_1, f_1) = \omega(e_2, f_2) = 1$$

Under this basis, $\omega(x, y) = x^t J y$, where

$$J = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}$$

A symplectomorphism is just a matrix satisfying $T^\omega(x, y) = \omega(Tx, Ty) = xT^tJTy = x^tJy$, i.e.,

$T^tJT = J$, and we get by doing calculus on $\mathrm{GL}_4(\mathbb{C})$,

$$\mathfrak{sp}(V) \cong \mathfrak{sp}_4\mathbb{C} = \{X \in \mathbb{C}^{2\times 2} | X^tJ + JX = 0\}$$

Let's now find a basis of $\mathfrak{sp}_4\mathbb{C}$. To begin with, let's assume $X = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$, where A, B, C, D are 2×2 matrices. Then the condition to be in the symplectic algebra becomes $X^tJ + JX = \begin{pmatrix} -C^t & A^t \\ -D^t & B^t \end{pmatrix} + \begin{pmatrix} C & D \\ -A & -B \end{pmatrix} = \begin{pmatrix} C - C^t & A^t + D \\ -A - D^t & B - B^t \end{pmatrix} = 0$, i.e., B, C are symmetric and $A^t = -D^t$. More explicitly,

$$\mathfrak{sp}_4\mathbb{C} = \{ \begin{bmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} & \begin{pmatrix} e & f \\ f & g \end{pmatrix} \\ \begin{pmatrix} h & i \\ i & j \end{pmatrix} & \begin{pmatrix} -a & -c \\ -b & -d \end{pmatrix} \end{bmatrix} | a, b, c, d, e, f, g, h, i, j \in \mathbb{C}\}$$

Let E_{ij} denote the matrix whose element in the ith row and jth column is 1 and 0 otherwise. An apparent basis of $\mathfrak{sp}_4\mathbb{C}$ consists of $H_1 := E_{11} - E_{33}, H_2 := E_{22} - E_{44}$, $X_{12} = E_{12} - E_{43}$, $X_{21} = E_{21} - E_{34}$, $Y_{14} = E_{14} - E_{23}$, $Y_{32} = E_{32} - E_{41}$, $Z_{13} = E_{13}$, $Z_{24} = E_{24}$, $Z_{31} = E_{31}$, $Z_{42} = E_{42}$. By comparing with the special linear algebras, we guess that $\mathfrak{h} = \mathrm{span}_{\mathbb{C}}\{H_1, H_2\}$ is the Cartan of $\mathfrak{sp}_4\mathbb{C}$. We carry over the argument to show:

Proposition 1.1. *$\mathfrak{h}$ is the Cartan of $\mathfrak{sp}_4\mathbb{C}$, i.e., $\mathfrak{h}$ is the maximal commuting subalgebra of $\mathfrak{sp}_4\mathbb{C}$ acting diagonally on $\mathfrak{sp}_4\mathbb{C}$ via adjoint representation.*

Proof. I will prove this by brutal force because this also gives us the roots.

$[H_1, H_2] = [E_{11} - E_{33}, E_{22} - E_{44}] = 0$, so $\mathfrak{h}$ is abelian.

$[H_1, X_{12}] = [E_{11} - E_{33}, E_{12} - E_{43}] = E_{12} - E_{43}$ and $[H_2, X_{12}] = [E_{22} - E_{44}, E_{12} - E_{43}] = -E_{12} + E_{43}$.

$[H_1, X_{21}] = [E_{11} - E_{33}, E_{21} - E_{34}] = -E_{21} + E_{34}$ and $[H_2, X_{21}] = [E_{22} - E_{44}, E_{21} - E_{34}] = E_{21} - E_{34}$.

$[H_1, Y_{14}] = [E_{11} - E_{33}, E_{14} - E_{23}] = E_{14} - E_{23}$ and $[H_2, Y_{14}] = [E_{22} - E_{44}, E_{14} - E_{23}] = -E_{23} + E_{14}$.

$[H_1, Y_{32}] = [E_{11} - E_{33}, E_{32} - E_{41}] = -E_{32} + E_{41}$ and $[H_2, Y_{32}] = [E_{22} - E_{44}, E_{32} - E_{41}] = -E_{32} + E_{44}$.

$[H_1, Z_{13}] = [E_{11} - E_{33}, E_{13}] = 2E_{13}$ and $[H_2, Z_{13}] = [E_{22} - E_{44}, E_{13}] = 0$.

$[H_1, Z_{24}] = [E_{11} - E_{33}, E_{24}] = 0$ and $[H_2, Z_{24}] = [E_{22} - E_{44}, E_{24}] = 2E_{24}$.

$[H_1, Z_{31}] = [E_{11} - E_{33}, E_{31}] = -2E_{31}$ and $[H_2, Z_{31}] = [E_{22} - E_{44}, E_{31}] = 0$.

$[H_1, Z_{42}] = [E_{11} - E_{33}, E_{42}] = 0$ and $[H_2, Z_{42}] = [E_{22} - E_{44}, E_{42}] = -2E_{42}$.

By the computation above, $\mathfrak{h}$ acts diagonally on $\mathfrak{sp}_4\mathbb{C}$. $\mathfrak{h}$ is also maximal because no other basis vectors are killed by $\mathfrak{h}$. $\square$

Now denote $L_1 = H_1^*$ and $L_2 = H_2^*$. Then $\mathfrak{h}^* = \mathrm{span}_{\mathbb{C}}\{L_1, L_2\}$. From the computation above, we have

Proposition 1.2. *Roots of $\mathfrak{sp}_4\mathbb{C}$ are $\pm 2L_1, \pm 2L_2, \pm L_1 \pm L_2$. So the roots of $\mathfrak{sp}_4\mathbb{C}$ in the weight diagram should look like*

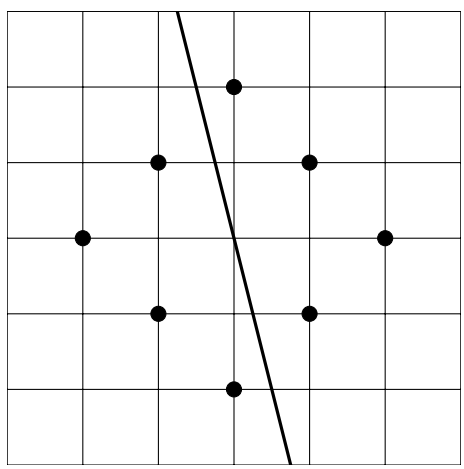

Figure 1: Roots of $\mathfrak{sp}_4\mathbb{C}$.

There are two things hidden in the above weight diagram: 1. L_1 is perpendicular to L_2; 2. the weight lattice Λ_W is indeed the intersection points formed by the grid.

To prove 1, we compute its Killing form κ:

Proposition 1.3. *Under κ, L_1 is perpendicular to L_2. In particular, the angle in the above weight lattice diagram is correct.*

Proof. By definition, $\kappa(H_i, H_j) = Tr(\mathrm{ad}H_i\mathrm{ad}H_j)$. Since $\mathfrak{sp}_4\mathbb{C} = \mathfrak{h} \bigoplus \mathfrak{sp}_4\mathbb{C}_\alpha$, where α's are the roots, $\mathrm{ad}H_i\mathrm{ad}H_j(v_\alpha) = \alpha(H_i)\alpha(H_j)v_\alpha$, so $Tr(\mathrm{ad}H_i\mathrm{ad}H_j) = \sum \alpha(H_i)\alpha(H_j)$. So by proposition 1.2, $\kappa(H_1, H_1) = \kappa(H_2, H_2) = 12$, $\kappa(H_1, H_2) = 0$. So this also holds in the dual space. $\square$

To prove 2, we locate copies of $\mathfrak{sl}_2\mathbb{C}$:

Proposition 1.4. *The weight lattice $\Lambda_W = \mathrm{span}_{\mathbb{Z}}\{L_1, L_2\}$.*

Proof. Let α be a weight of an irreducible representation V and β be a root of $\mathfrak{sp}_4\mathbb{C}$, then $\bigoplus V_{\alpha+k\beta}$ is an $\mathfrak{sl}_2\mathbb{C}$-subrepresentation: let h, e, f the elements as in our class, let $w_\beta \in \mathfrak{sp}_4\mathbb{C}_\beta$ and $v_\alpha \in V_\alpha$. Then $e \cdot v_\alpha = w_\beta \cdot v_\alpha$, $f \cdot v_\alpha = w_{-\beta} \cdot v_\alpha$, so $[e, f] \cdot v_\alpha = h \cdot v_\alpha = [w_\beta, w_{-\beta}] \cdot v_\alpha$. Let me compute all the cases:

1. $\beta = L_1 - L_2$: $[X_{12}, X_{21}] = [E_{12} - E_{43}, E_{21} - E_{34}] = E_{22} - E_{44} + E_{11} + E_{44} = H_1 + H_2$. In this case, the result of $\mathfrak{sl}_2\mathbb{C}$ tells us that $h \cdot v_\alpha = H_2 \cdot v_\alpha = \alpha(H_1 + H_2)v_\alpha$, i.e., $\alpha(H_1) + \alpha(H_2) \in \mathbb{Z}$.
2. $\beta = 2L_1$: $[Z_{13}, Z_{31}] = [E_{13}, E_{31}] = E_{11} - E_{33} = H_1$, so $H_1 \cdot v_\alpha$ and so $\alpha(H_1) \in \mathbb{Z}$.
3. $\beta = 2L_2$: $[Z_{24}, Z_{42}] = H_2$, so $\alpha(H_2) \in \mathbb{Z}$.
4. $\beta = L_1 + L_2$: $[Y_{14}, Y_{32}] = [E_{14} - E_{23}, E_{32} - E_{41}] = -E_{11} - E_{22} + E_{44} + E_{33} = -H_1 - H_2$, so $-\alpha(H_1) - \alpha(H_2) \in \mathbb{Z}$.

Now let $aL_1 + bL_2 \in \mathfrak{h}^*$, then the above conditions force $a, b \in \mathbb{Z}$ and $a + b \in \mathbb{Z}$, so $a, b \in \mathbb{Z}$, and the weight lattice $\Lambda_W \subseteq \{aL_1 + bL_2 | a, b \in \mathbb{Z}\}$. To show the other direction of inclusion, we need to construct some representations, and I'll assume it's true as it'll be clear as the paper goes. $\square$

The last thing to do is to choose a positive direction, let's call the roots on the right of the bold line positive and those on the left negative. The simple roots are then $\{2L_2, L_1 - L_2\}$. The Weyl chamber is then the cone with rays generated by $2L_1$ and $L_1 + L_2$. So for any representation, its highest weights will be non-negative linear combinations of $L_1, L_1 + L_2$, i.e., in the form $(a+b)L_1 + bL_2$. Denote $\Gamma_{a,b}$ to be the unique irreducible representation with the highest weight $(a+b)L_1 + bL_2$.

This finishes the setup of weight diagrams.

2 Representations of $\mathfrak{sp}_4\mathbb{C}$

2.1 The exterior powers $\bigwedge^n V$

If $n = 1$, by direct computation, e_i has weight L_i, f_i has weight $-L_i$. So the weight diagram is as in figure 2. In particular, since the highest weight vector is L_1, we denote this by $\Gamma_{1,0}$. I'll use these two notations interchangeably.

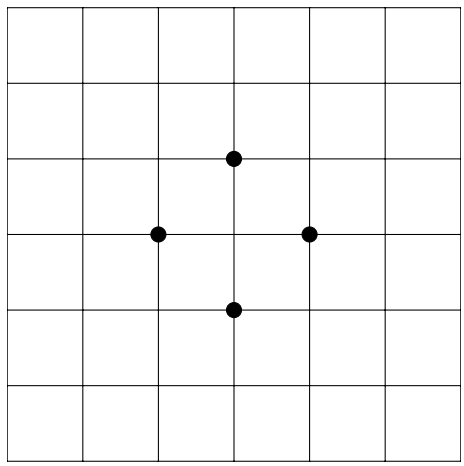

Figure 2: Roots of $\mathfrak{sp}_4\mathbb{C}$

If $n = 2$, the weights of $\Lambda^2 V$ is the pairwise sum of different weights of V, which are $\pm L_1 \pm L_2$ and 0 with multiplicity 2 as in figure 3. This contains an irreducible subrepresentation generated by the highest weight vector $L_1 + L_2$, denoted by $\Gamma_{0,1}$. Taking $\Gamma_{0,1}$ away we have a copy of $\mathbb{C}$, so the conclusion is $\bigwedge^2 V \cong \Gamma_{0,1} \oplus \mathbb{C}$.

$\Lambda^3 V, \Lambda^4 V$ are isomorphic to the previous two cases because of the dimension reason.

2.2 The symmetric powers $\mathrm{Sym}^2\Gamma_{1,0}$ and $\mathrm{Sym}^2\Gamma_{0,1}$

Weights of $\mathrm{Sym}^2 V$ are the pairwise sum of weights of V, so weights are $\pm 2L_1, \pm 2L_2, \pm L_1 \pm L_2$ and 0 with multiplicity 2 as in figure 4. The irreducible subrepresentation generated by the highest weight

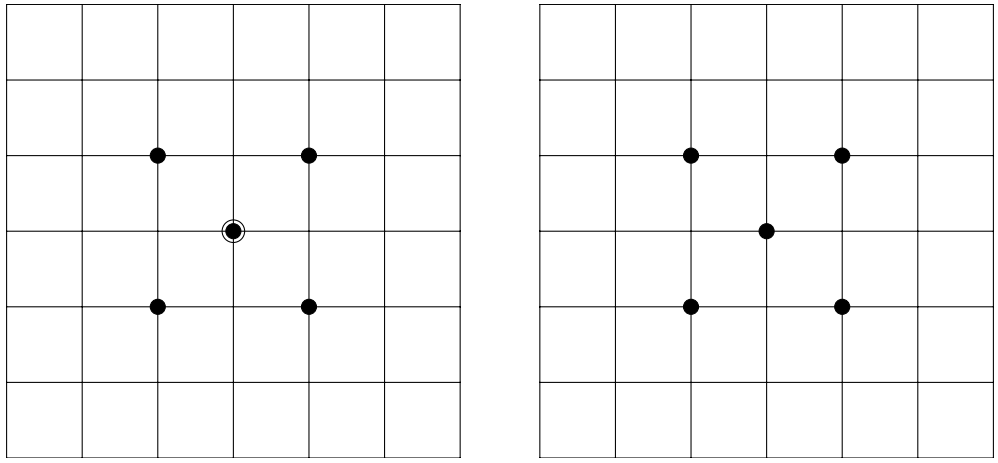

Figure 3: Left: Weight diagram of $\bigwedge^2 V$; Right: Weight diagram of $\Gamma_{0,1}$

vector $2L_1$ is isomorphic to the adjoint representation $\mathfrak{sp}_4\mathbb{C}$. The question now is to determine if it's the whole $\mathrm{Sym}^2 V$. There are two ways to go to 0: 1. start from $L_1 + L_2$ and apply a $g_{-L_1-L_2}$, or 2. start from $2L_1$ and apply g_{-2L_1}. In case 1, an eigenvector of weight $L_1 + L_2$ is e_1e_2, and apply $Y_{32} \cdot e_1e_2 = (Y \cdot e_1)e_2 + e_1(Y \cdot e_2) = ((E_{32} - E_{41})e_1)e_2 + e_1((E_{32} - E_{41})e_2) = -e_4e_2 + e_1e_3$. In case 2, an eigenvector of weight $2L_1$ is e_1e_1, and apply $(E_{31}e_1)e_1 + e_1(E_{31}e_1) = 2e_1e_3$. Since $-e_4e_2 + e_1e_3$ and $2e_1e_3$ are linearly independent, the dimension of eigenspace of weight 0 is 2, so the multiplicity of 0 is 2. The conclusion here is that $\mathrm{Sym}^2 V \cong \mathfrak{sp}_4\mathbb{C}$ is a $\Gamma_{2,0}$.

Weights of $\mathrm{Sym}^2\Gamma_{0,1}$ are also pairwise sum of weights of $\Gamma_{0,1}$, so the weight diagram is as in figure 4 right. The highest weight vector in this case is $2L_1 + 2L_2$. The claim is that this contains an irreducible $\Gamma_{0,2}$ with an additional multiplicity in 0, i.e., $\mathrm{Sym}^2\Gamma_{0,1} \cong \Gamma_{0,2} \oplus \mathbb{C}$. The way to show this is again by brutal computation.

2.3 The representation $\Gamma_{1,0} \otimes \Gamma_{0,1}$

In $\Gamma_{1,0} \otimes \Gamma_{0,1}$, we expect to get a $\Gamma_{1,1}$. To start with, weights of $\Gamma_{1,0} \otimes \Gamma_{0,1}$ are sums of weights of $\Gamma_{1,0}$ and $\Gamma_{0,1}$ as in figure 5 left. The highest weight vector is $L_{1,1}$. And the claim here is that $\Gamma_{1,0} \otimes \Gamma_{0,1} \cong \Gamma_{1,1} \oplus \Gamma_{0,1}$. So the weight diagram of $\Gamma_{1,1}$ is as in figure 5 right.

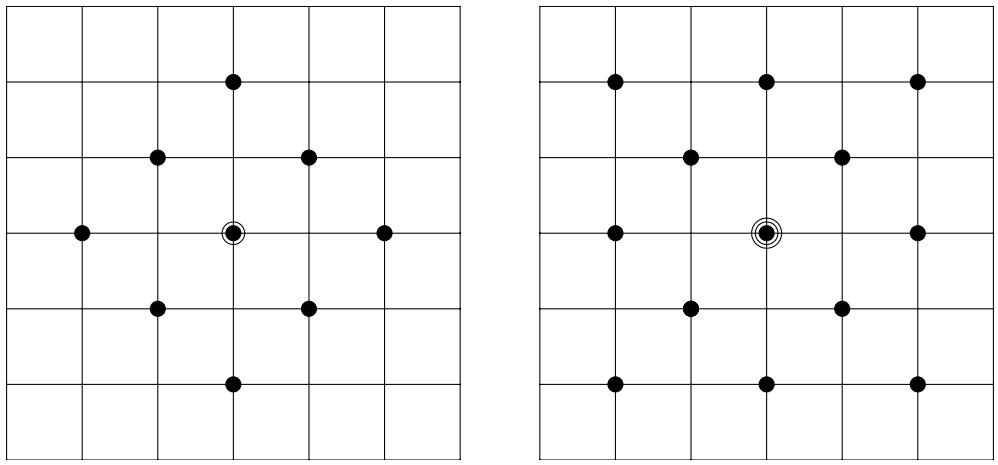

Figure 4: Left: Weight diagram of $\mathrm{Sym}^2\Gamma_{1,0} =: \Gamma_{2,0}$. Right: Weight diagram of $\mathrm{Sym}^2\Gamma_{0,1} = \Gamma_{0,2} \oplus \mathbb{C}$

3 General patterns for $\mathfrak{sp}_4\mathbb{C}$-representations

3.1 $\Gamma_{n,0}$

The cases $n = 1, 2$ are listed in figure 2 and 4. For $n \geq 3$, the idea is to show $\mathrm{Sym}^n\Gamma_{1,0}$ is irreducible, then because it has to contain a highest weight vector of weight nL_1, $\mathrm{Sym}^n\Gamma_{1,0} = \Gamma_{n,0}$.

Proposition 3.1. $\mathrm{Sym}^n\Gamma_{1,0}$ *is irreducible. Moreover, the weight diagram of* $\Gamma_{n,0}$ *consists of diamonds with vertices* $nL_1, (n-2a)L_1, \ldots$. *Let us denote* D_{a+1} *to be the diamond with vertex* $(n-2a)L_1$. *Then each weight in* D_{a+1} *has multiplicity* $a+1$.

Proof. A convenient way to see $\mathrm{Sym}^n\Gamma_{1,0}$ is irreducible is to use Weyl's dimension formula to show the dimension of $\Gamma_{n,0}$ equals the dimension of $\mathrm{Sym}^n\Gamma$.

For the weight diagram, we know that weights of $\mathrm{Sym}^n\Gamma_{1,0}$ are sums of n roots of $\Gamma_{1,0}$, i.e., sums of n elements from $\{\pm L_1, \pm L_2\}$. Let's first focus on the first quadrant:

1. $\mathrm{Sym}^n\Gamma_{1,0}$ can have weight nL_1 with multiplicity 1, which is when we choose all n elements to be L_1 and sum them, which has in total 1 choice. More generally, it can have weight $(n-2a)L_1$ as long as $n \geq 2a$ because we can choose up to a pairs of L_2 in the summand, giving us multiplicity $a+1$. Exactly the same argument tells us that $(n-2a)L_2$ has multiplicty $a+1$.

2. Connect $(n-2a)L_1$ and $(n-2a)L_2$ by a line segment, the claim is that each integer point

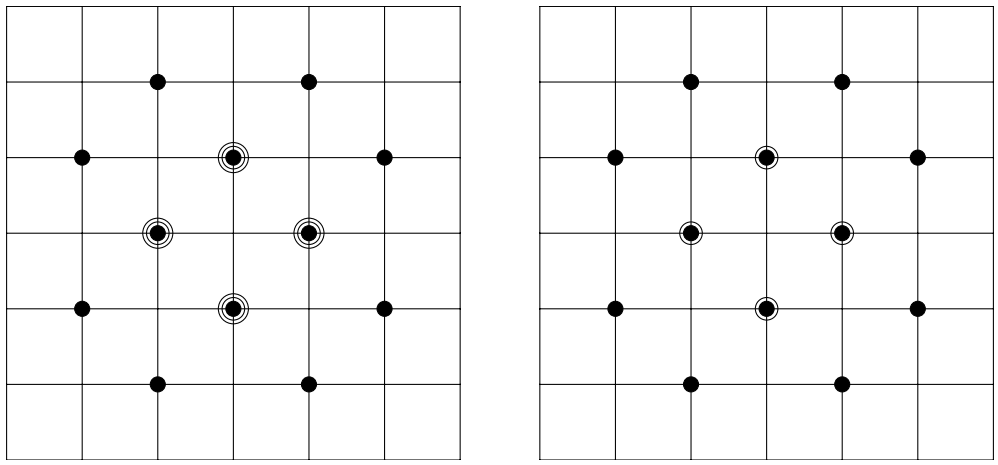

Figure 5: Left: Weight diagram of $\Gamma_{1,0} \otimes \Gamma_{0,1}$. Right: Weight diagram of $\Gamma_{1,1}$

passed by this line is a weight with multiplicity $a+1$. To see this, note that this line is defined by $x+y=n-2a$, so points on this line are of the form $bL_1+(n-2a-b)L_2$. So we need b many L_1 and $(n-2a-b)L_2$, and are free to choose the number of pairs of L_1 in the other a terms, which gives us again multiplicity $a+1$.

3. Points outside the furthest diamond D_1 are not weights because the highest weight is nL_1. Points inside D_1 but not on these diamonds are not weights because of the parity of n doesn't allow this to happen. For example, $(n-1)L_1$ is never a weight because n has different parity than $n-1$. So the weight diagram described in the proposition is the correct one. □

3.2 $\Gamma_{0,n}$

Here I'll claim the following pattern of $\Gamma_{0,n}$ representations. For a proof, see [2].

Proposition 3.2. *Weight diagrams of $\Gamma_{0,n}$ consists of n squares centered at the origin with vertices $0, L_1+L_2, ..., n(L_1+L_2)$. Each square has a specific multiplicity which follows the pattern $1,1,3,3,6,6,10,10,...$, starting with the furthest square and moving to the nearest.*

3.3 $\Gamma_{m,n}$

Weight diagrams of $\Gamma_{m,n}$ has a pattern that is harder to compute, see [1].

4 Conclusion

Now weight diagrams of all irreducible representations has been clearly computed. It turns out that given any weight diagram of $\mathfrak{sp}_4\mathbb{C}$ representations, we can mimic what we did for $\mathfrak{sl}_n\mathbb{C}$ in class, find the direct summands based on what we've found. By the clarity of this procedure, I think it's possible to write a program with input the weight diagram and output the direct sum decomposition of the representation.

References

[1] Julia Maddox. An elementary approach to the weight multiplicities for $\mathfrak{sp}_4\mathbb{C}$. https://math.ou.edu/ jmaddox/jmaddox.SP4.pdf, 2012.

[2] Joe Harris William Fulton. *Representation Theory.* 2004.